adapted on
Lynford L. Goddard
Brian T. Cunningham

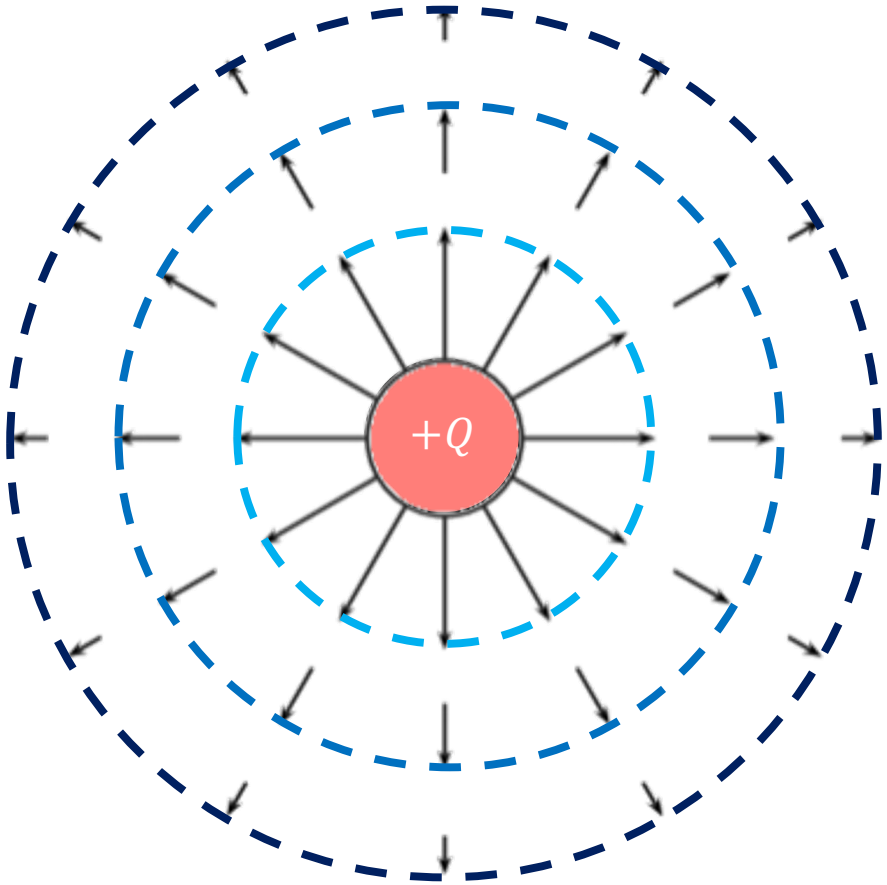
Lectures 3

Surface Integrals
Connecting Coulomb's and Gauss' Law

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Electric Field Around a Point Charge



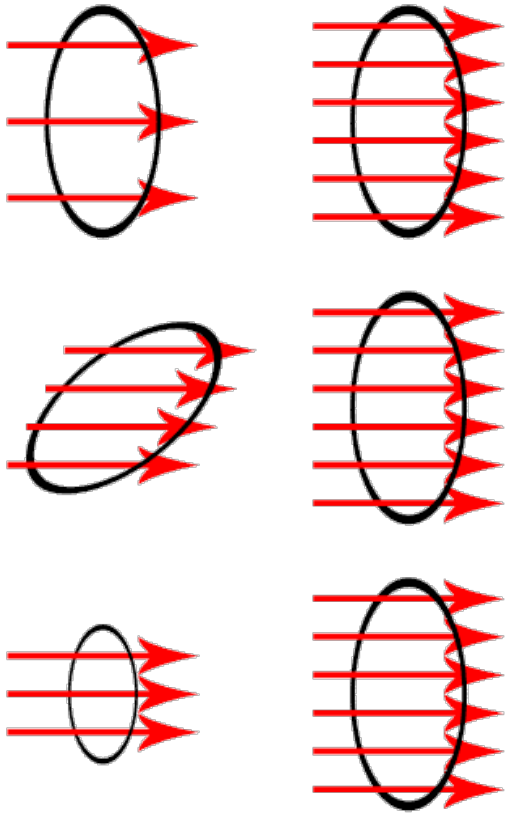
$$\vec{E} = \frac{Q}{4\pi\varepsilon_0 R^2}\,\hat{a}_R$$

Gauss' Law

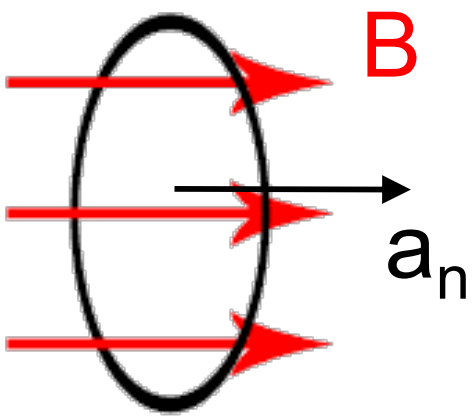
Number of field lines passing through **any** surface that encloses Q_1 is constant.

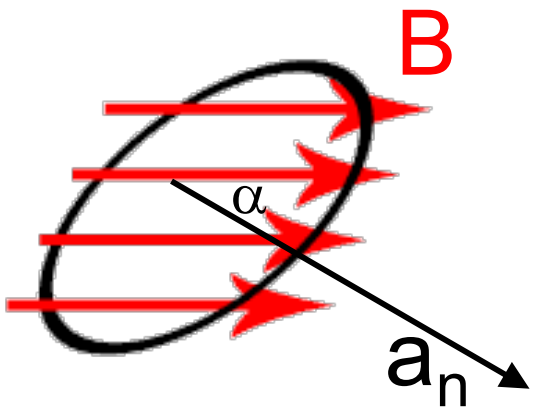
Surface Integrals

- Flux = # of arrows that pass thru a surface; it depends on:
  - The density of vectors
  - The angle of the surface
  - The area of the surface



Surface Integral describes the Flux of a Vector Field

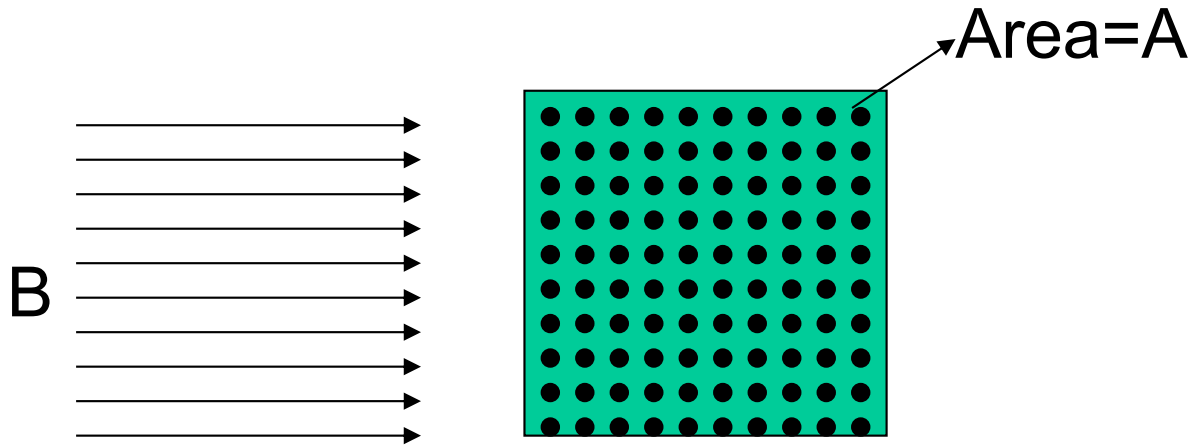




$$Flux = \vec{B} \bullet \Delta\vec{S}$$
$$= (\vec{B} \bullet \hat{a}_n)\left|\Delta S\right|$$
$$= (B\cos\alpha)\left|\Delta S\right|$$
$$= \left|B_n\right|\left|\Delta S\right|$$

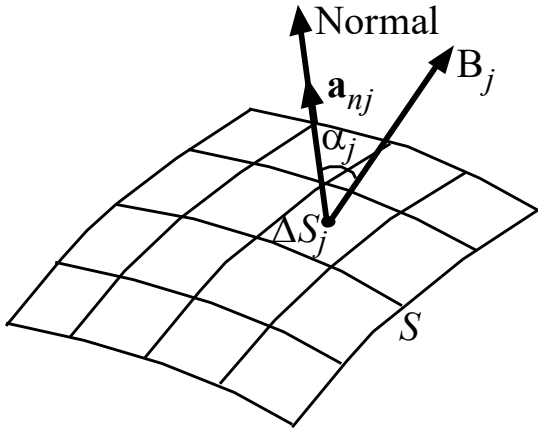
Trick that works sometimes

If the flux is
- Uniform (has equal magnitude across whole surface)
- Perpendicular to the surface



$$\psi = \iint_S \vec{B} \bullet d\vec{S} = B \cdot A$$

In the general case …



$$\text{Flux} = \sum_{j=1}^{n} \Delta\psi_j$$

$$= \sum_{j=1}^{n} \mathbf{B}_j \bullet \Delta\mathbf{S}_j$$

In the limit $n \to \infty$,

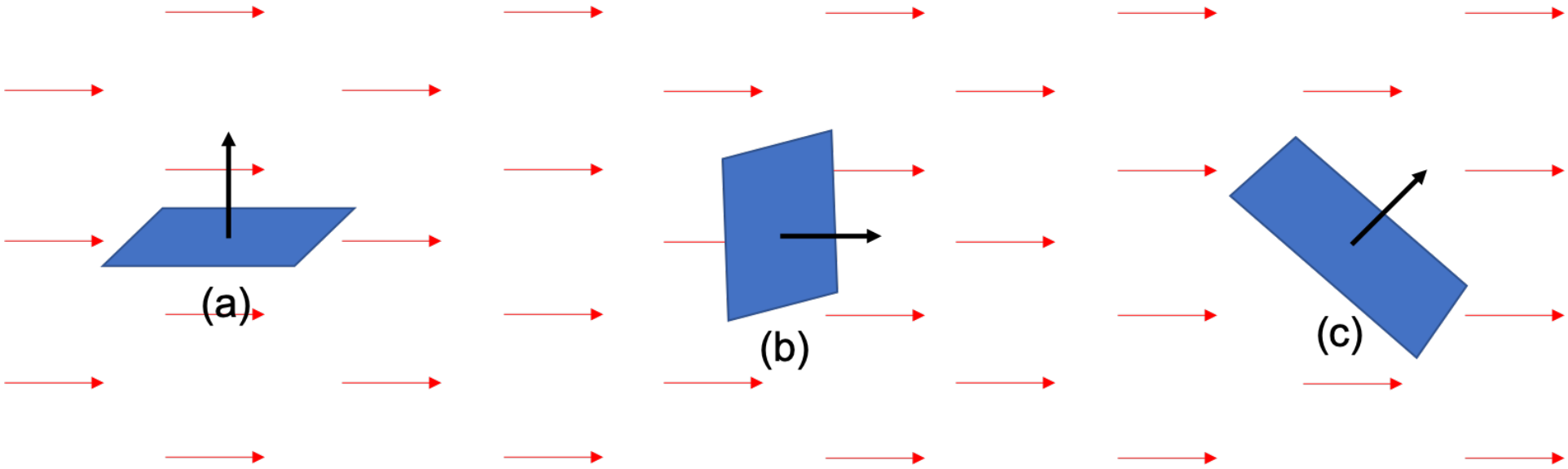
$$\text{Flux, } \boxed{\psi = \int_S \mathbf{B} \bullet d\mathbf{S}}$$

= Surface integral of B over S.

Q:

Which of the following three case has more flux?

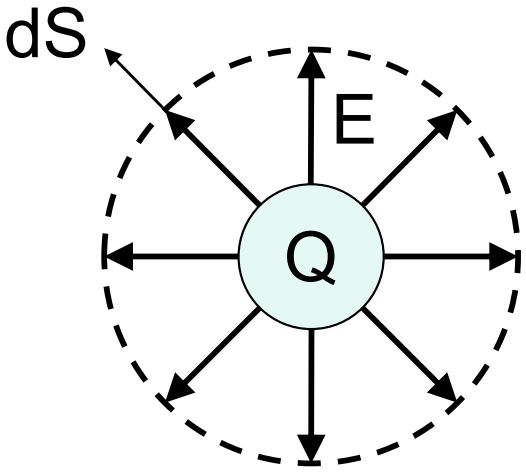
Compare $\vec{E} \cdot \vec{A}$



Electric Flux = Charge Enclosed

- Coulomb's Law for the Electric field of a point charge:

$$\vec{E} = \frac{Q}{4\pi\varepsilon_0 R^2}\hat{a}_R$$



Define **D=ε_0E** to be the "displacement flux density"

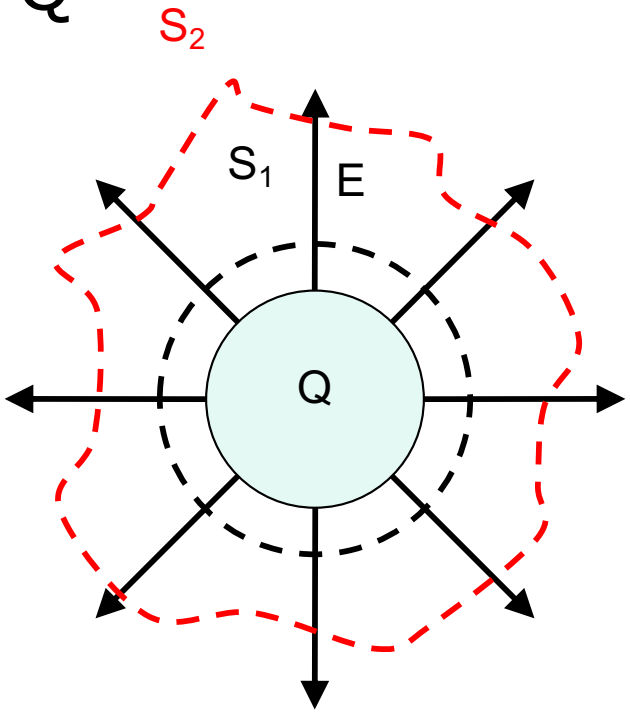
$$\psi_E = \oiint_S \vec{D} \bullet d\vec{S} = \oiint_S \varepsilon_0 \vec{E} \bullet d\vec{S}$$

$$= \varepsilon_0 E(Surf\ Area)$$

$$= \varepsilon_0 \frac{Q}{4\pi\varepsilon_0 R^2}(4\pi R^2) = Q$$

Same Flux Out of Any Surface

- Same # of field lines pass thru any surface that encloses Q

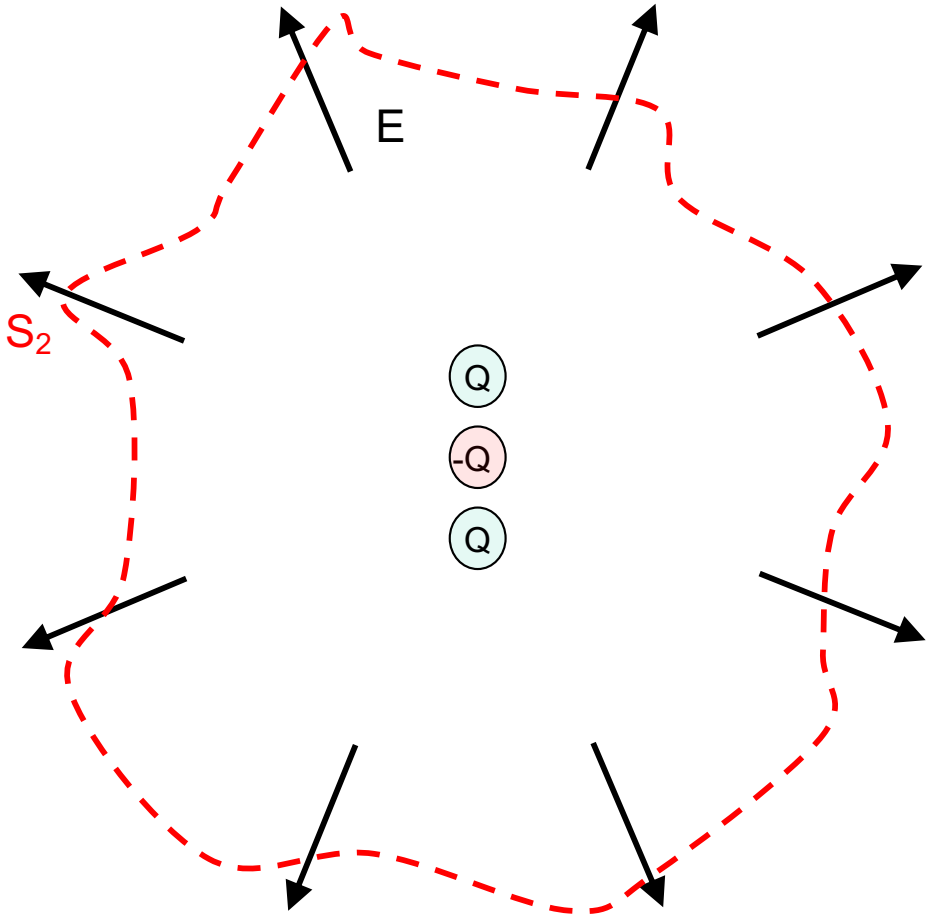


$$\psi_E = \oiint_S \vec{D} \bullet d\vec{S} = Q_{enclosed}$$

Our first Maxwell Equation!

Superposition

- The flux for an arbitrary distribution of charges is obtained using superposition

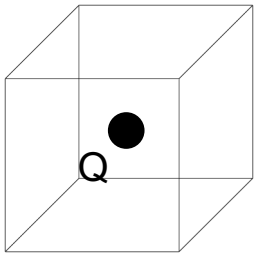


$$\psi_E = \oiint_S \vec{D} \bullet d\vec{S} = Q_{enclosed}$$

FLUX OUT = CHARGE ENCLOSED

Simple Example 1

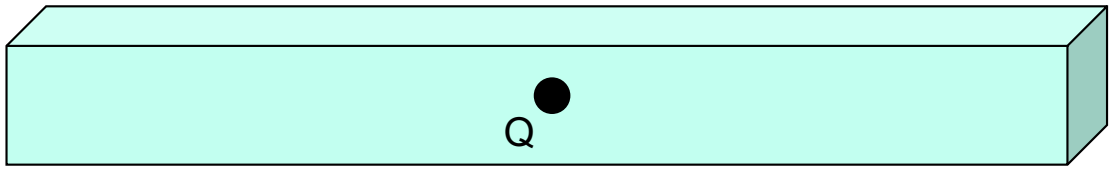
6-sided cube with Q at the center:



Flux out of entire box = Q
Flux out of one side = Q/6

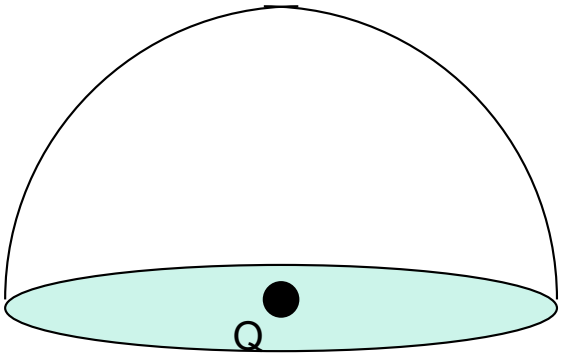
Simple Example 2

Flux pointing out of thin box on bottom surface with Q at the center



Simple Example 3

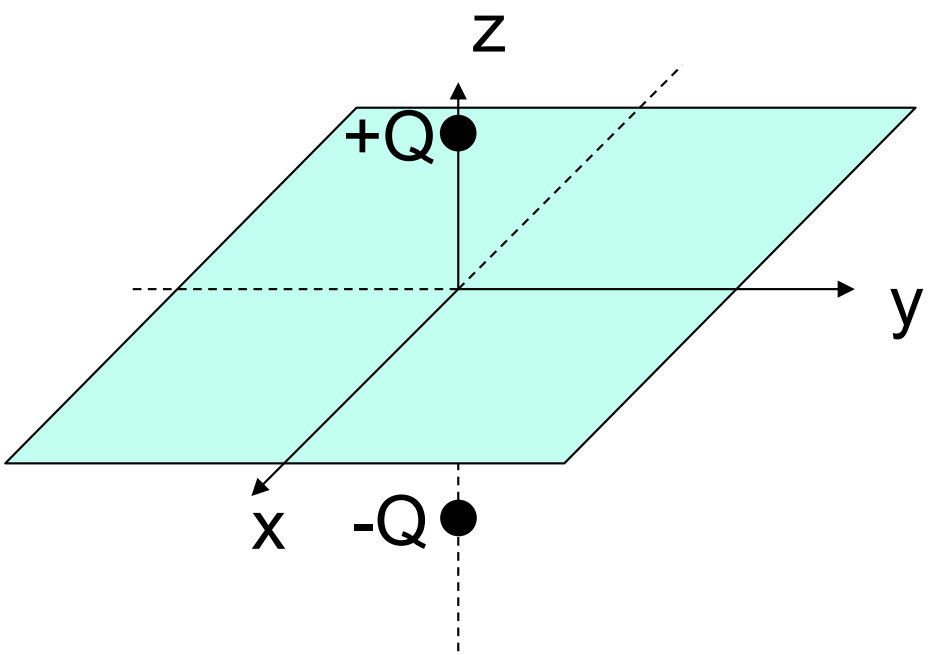
Flux out of hemisphere with Q at the center =



Challenge Question 1

- What is the flux across the xy plane $\mathbf{a}_n=\mathbf{a}_z$ for a dipole: +Q at (0,0,a/2) and -Q at (0,0,-a/2)?

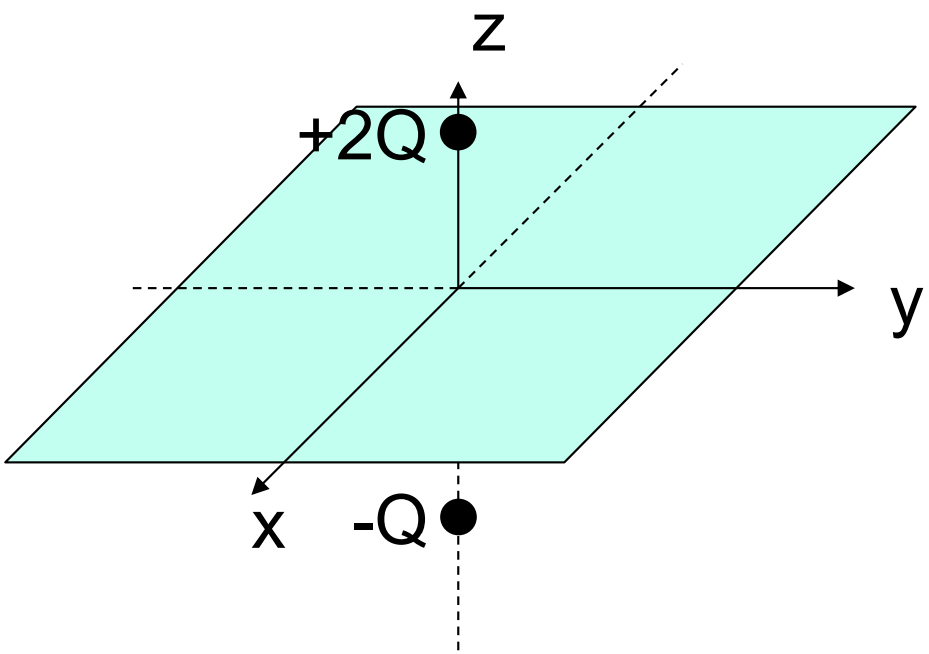
(a) Q/2
(b) -Q/2
(c) -Q
(d) -3Q/2
(e) -2Q



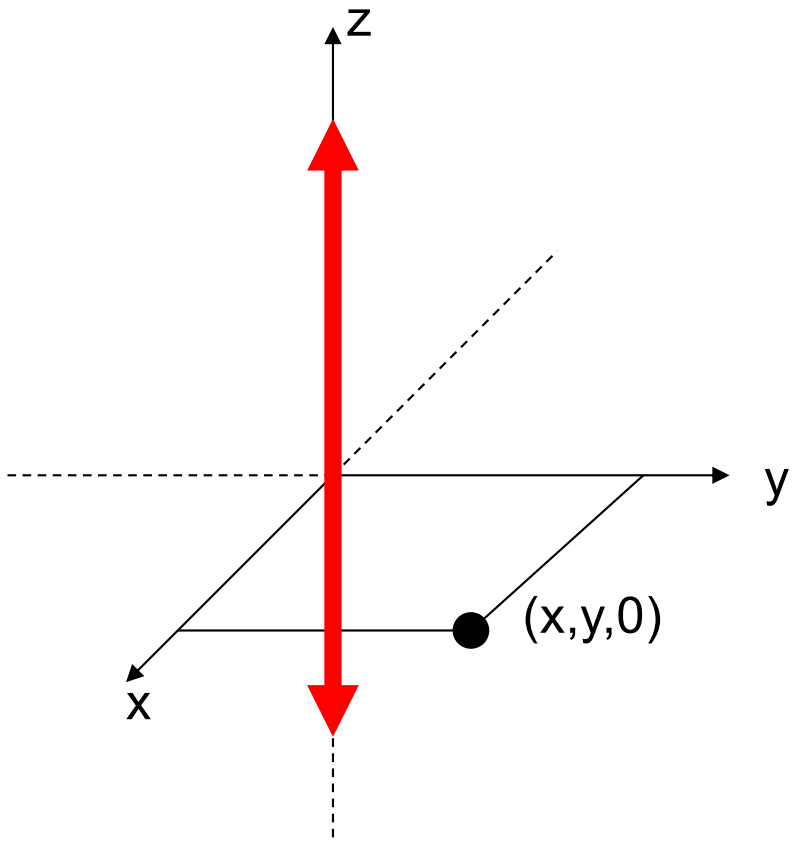
Challenge Question 2

- What is the flux across the xy plane $\mathbf{a}_n=\mathbf{a}_z$ for the charge distribution: +2Q at (0,0,a/2) and -Q at (0,0,-a/2)? (Hint: Use superposition.)

(a) Q/2
(b) -Q/2
(c) -Q
(d) -3Q/2
(e) -2Q



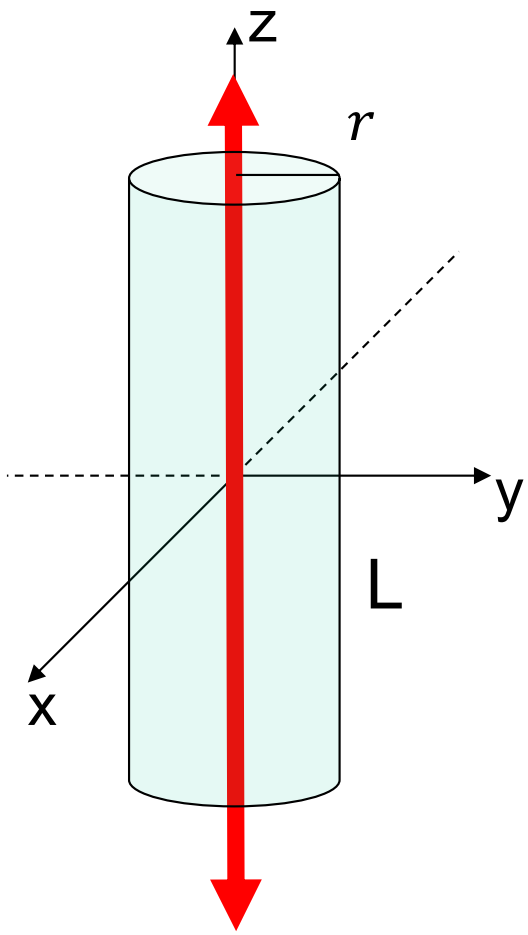
Example 3: **E** due to ∞ line of charge



Linear charge distribution ρ_l (C/m)

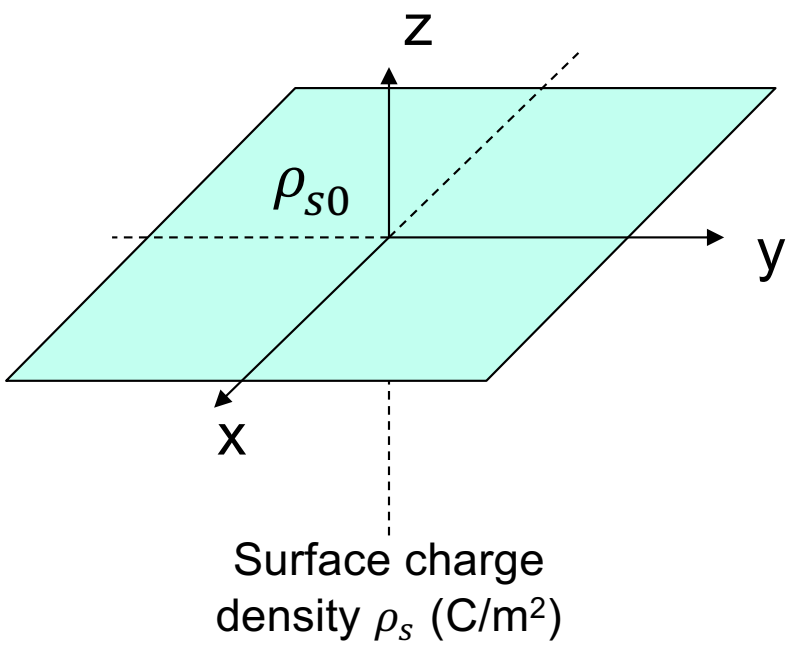
Find **E** at point on xy plane

Example 3: **E** due to ∞ line of charge



Gaussian pillbox

Example 4: **E** due to a surface of charge

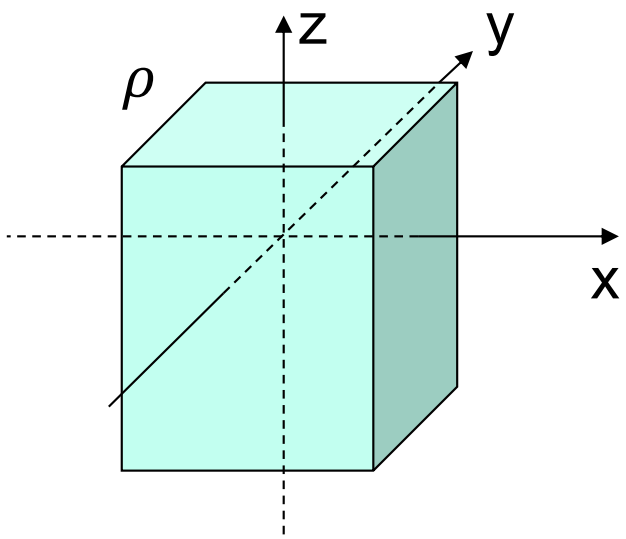


Point charges Q are distributed over z = 0 plane with an average surface charge density of ρ_{s0} C/m^2. Find **E**.

(Review at home)

Prof. Kudeki's lect03.pdf
Example 1

Example 5: **E** due to a slab of charge



Volume charge density ρ (C/m^3)

Point charges Q are distributed throughout an infinite slab of width W located over $-\frac{W}{2} < x < \frac{W}{2}$. Find **E**.

(Review at home)

Prof. Kudeki's lect03.pdf
Example 2

Summary of electrostatic fields E

- Coulomb field

$$\mathbf{E} = \hat{r}\frac{Q}{4\pi\epsilon_o r^2} \text{ of a point charge } Q,$$

- Field

$$\mathbf{E} = \hat{r}\frac{\lambda}{2\pi\epsilon_o r} \text{ of constant line density } \lambda,$$

- Field

$$\mathbf{E} = \hat{x}\frac{\rho_s}{2\epsilon_o}\text{sgn}(x) \text{ of constant surface density } \rho_s,$$

- Field

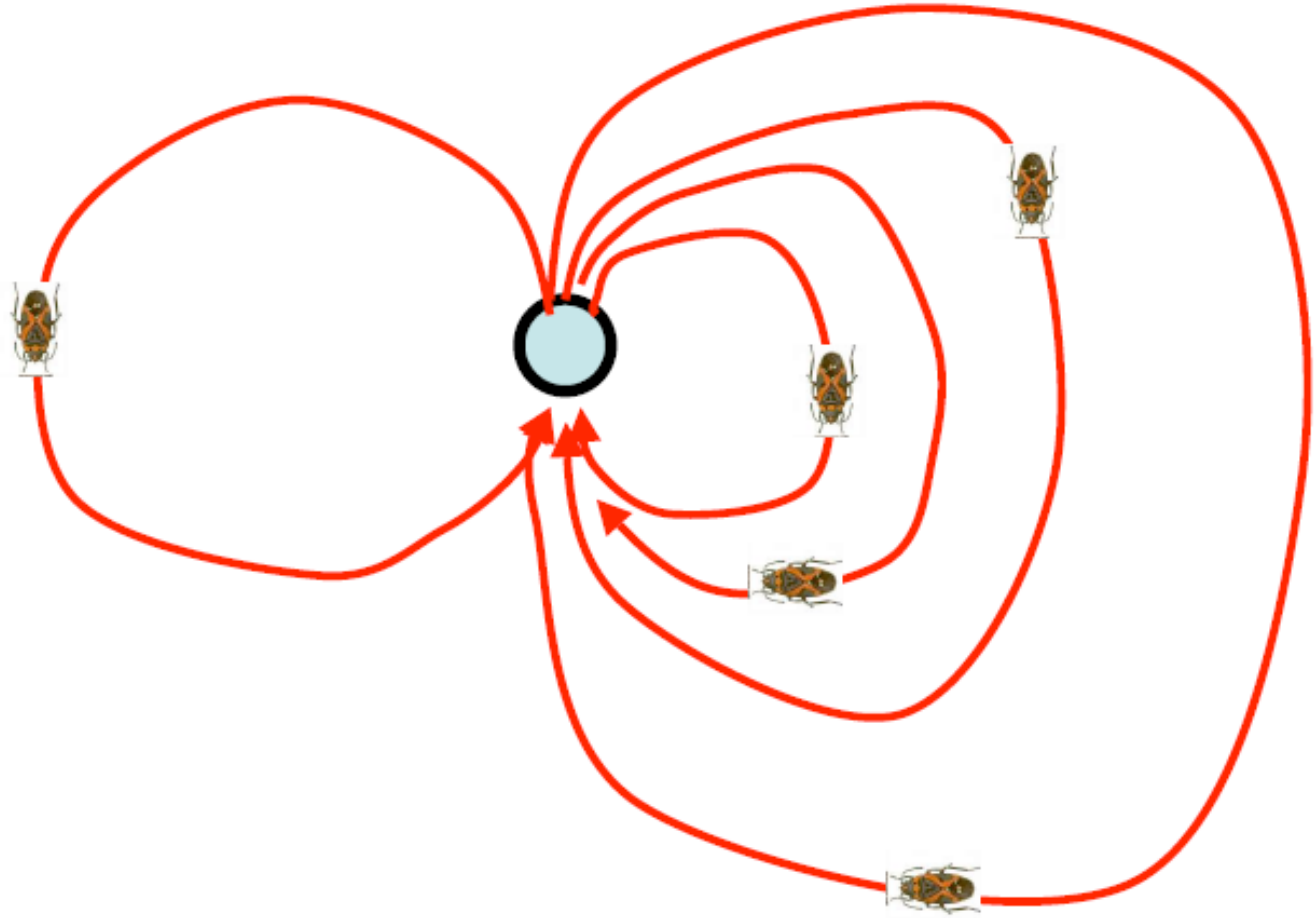
$$\mathbf{E} = \hat{x}\frac{\rho x}{\epsilon_o} \text{ of constant volume density } \rho$$

The (Fictional) Yadaraf Bug: Flux and Surface Integrals



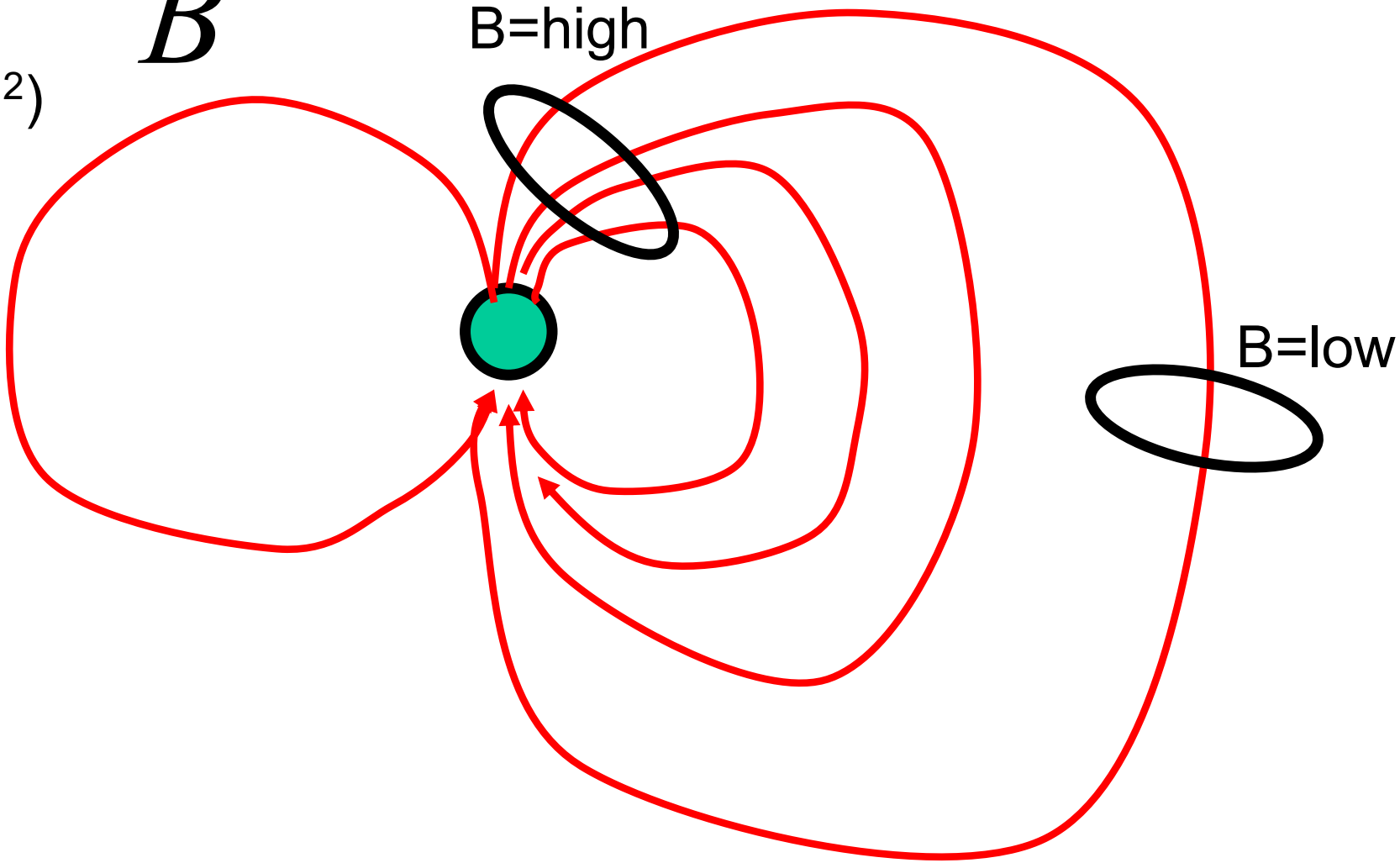
- The Yadaraf Bug
  - They live in the ground
  - They only come out at night to search for food
  - Very hard to see - REALLY small
  - After gathering food, they <u>always</u> return to their hole in the ground

Yadaraf Bug Travel Paths

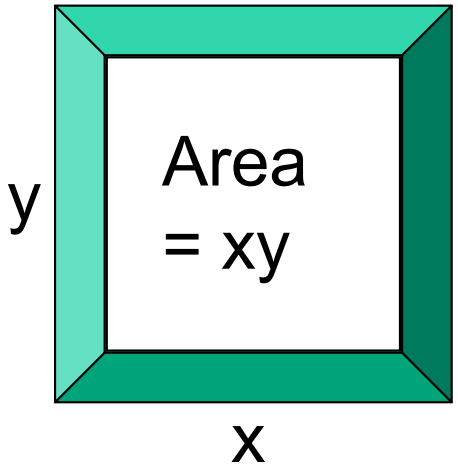


Yadaraf Bug Travel Paths

Bug Density Vector (Bugs/m^2) $\vec{B}$



Yadaraf Bug Counter

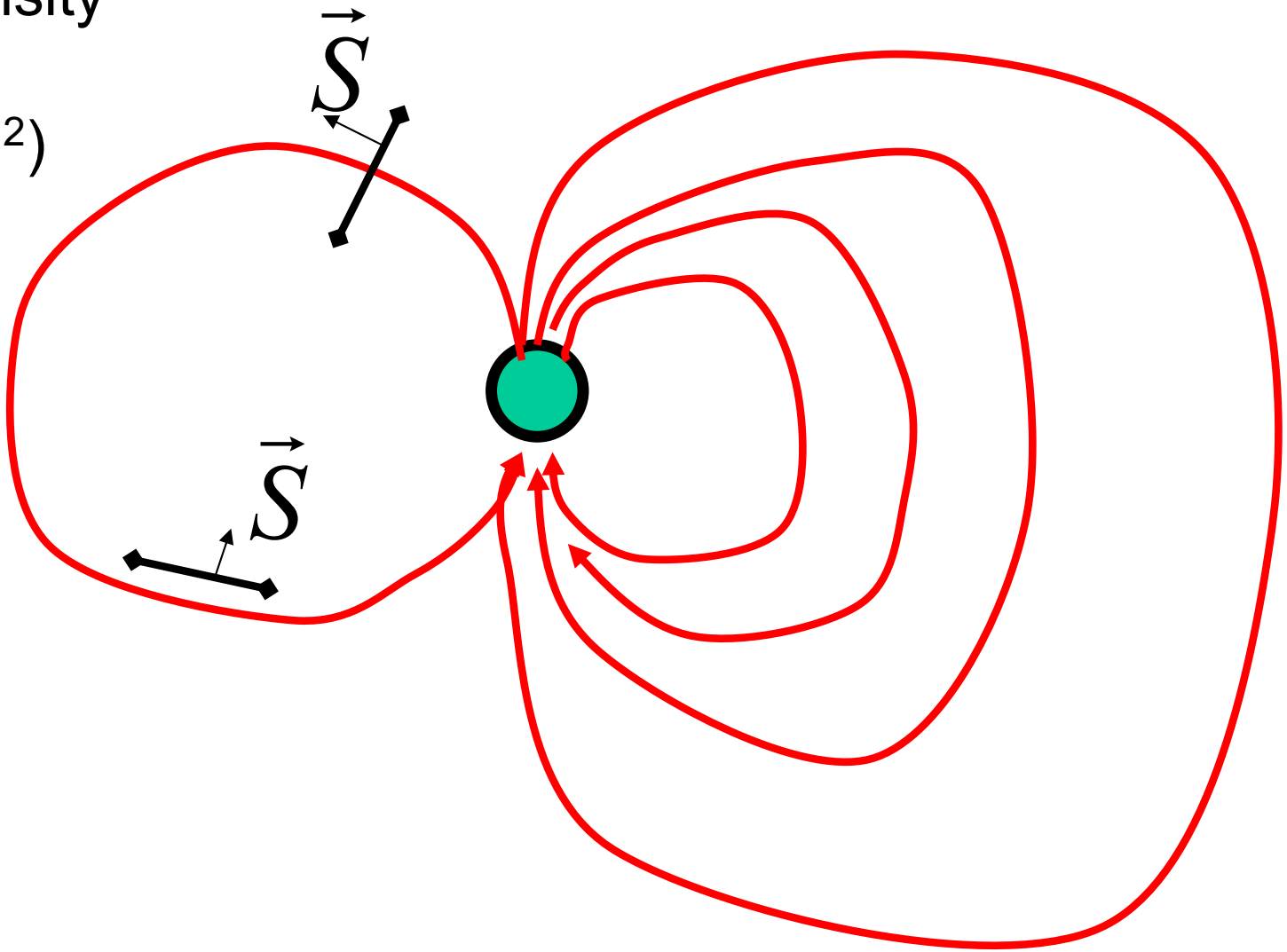


- Register a +1 count for each bug going through in one direction
- Register a -1 cound for each bug going through in the opposite direction
- Has a known area for the bugs to pass through

Yadaraf Bug Travel Paths

Bug Density Vectors (Bugs/m^2)

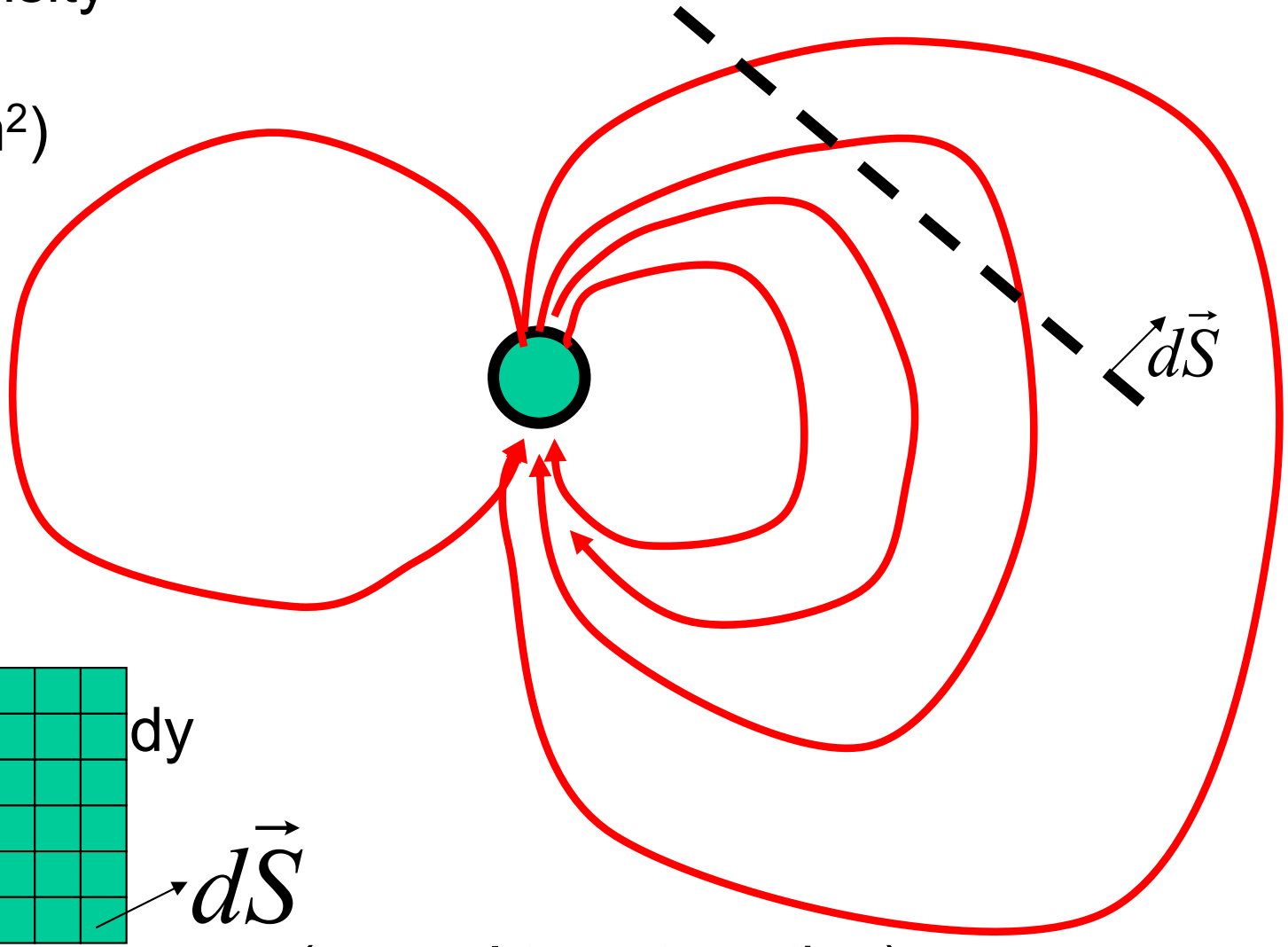
$\vec{B}$

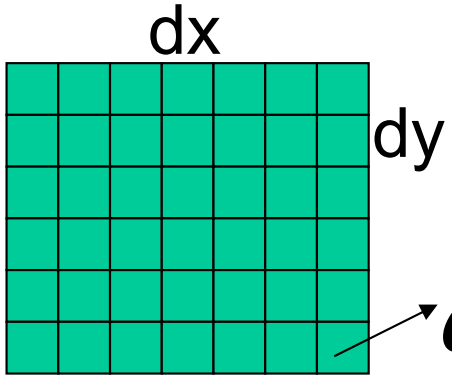


Bug Counting Net

Bug Density Vectors (Bugs/m^2)

$\vec{B}$



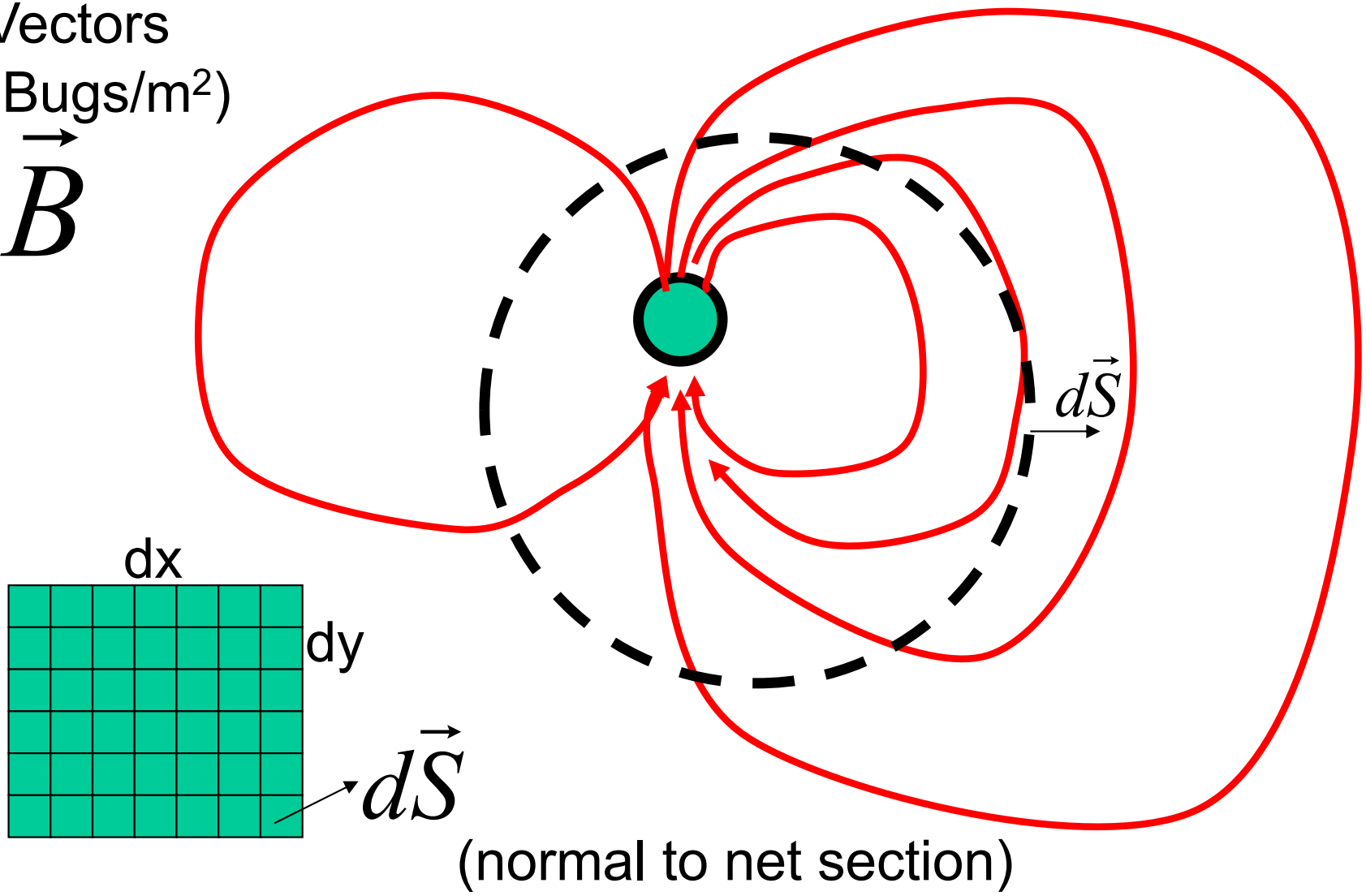


(normal to net section)

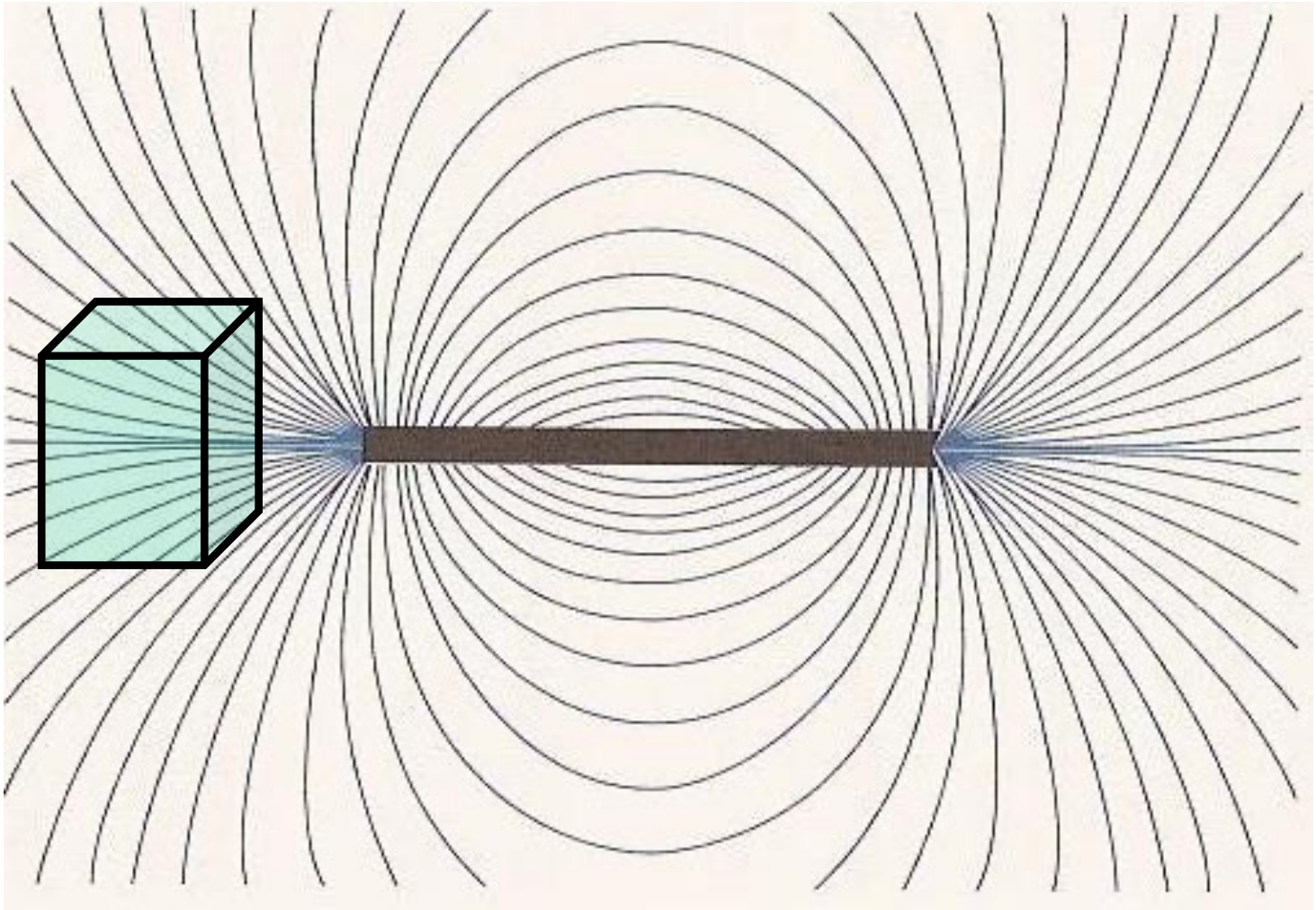
"Closed" Bug Counting Net

Bug Density Vectors (Bugs/m^2)

$\vec{B}$



So magnetic flux, ψ_B, through a CLOSED surface is always zero



Magnetic field lines form closed paths (they do not begin or end)

Gauss' Law for B Fields

Net flux of magnetic field lines through any closed surface MUST be zero.

$$\oiint_S \vec{B} \bullet d\vec{S} = 0$$

Our Second Maxwell Equation!

Lecture 3 Summary

- Gauss' Law

$$\psi_E = \oiint_S \vec{D} \bullet d\vec{S} = Q_{enclosed}$$

FLUX OUT = CHARGE ENCLOSED

- Review examples on sheet charges, slab charges (slides 23-26)

- Next lecture
  - Divergence and curl